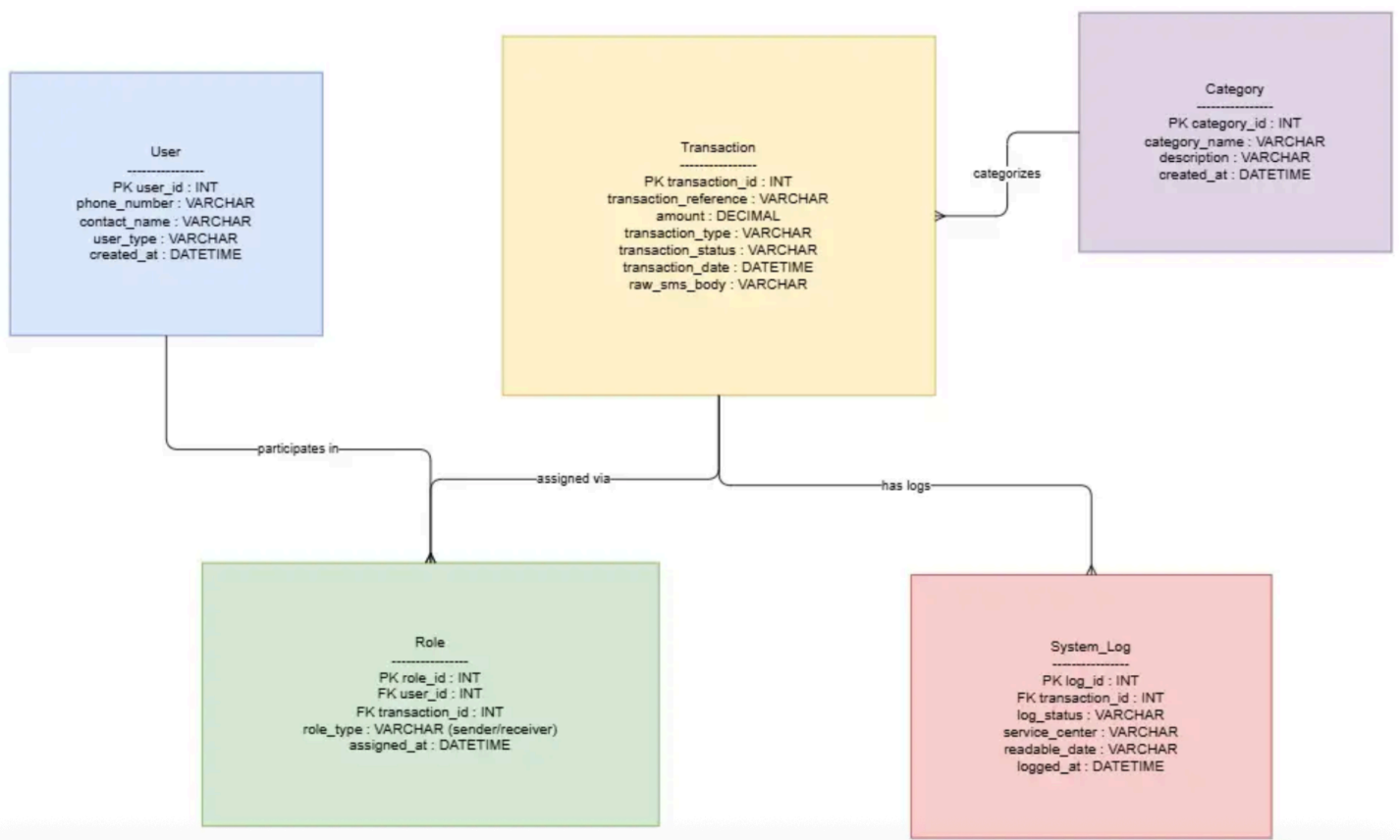


Database Design Document

1. Introduction

The MoMo SMS Data Processing System is a database solution designed to capture, store, and analyze mobile money transaction data extracted from SMS messages. Mobile money services enable users to send money, pay bills, purchase airtime, and conduct various financial transactions through their mobile phones. This database serves as the central repository for all transaction data, user information, and system logs, ensuring data integrity, enabling efficient querying, and supporting audit and compliance requirements. This document provides comprehensive documentation including the Entity Relationship Diagram, design rationale, data dictionary, sample queries, and security rules implemented to ensure data accuracy and protection.

2. ERD



3. Design Rationale and Justification

Entity Selection

The database schema consists of five interconnected entities designed to represent all aspects of MoMo SMS transaction data. The Transaction table is the central entity that stores core transaction details extracted from SMS messages including the transaction reference (TxId),

amount, type, status, date, and the original raw SMS body. The User table is a registry of all parties involved in transactions, identified by phone numbers and contact names extracted from the SMS data. The Category table provides a classification system for different transaction types identified in the XML data, such as money transfers, payments, bank deposits, airtime purchases, and data bundle purchases.

Relationship Design

The Role table functions as a junction table resolving the many-to-many relationship between Users and Transactions, as each transaction involves multiple users (sender and receiver) and each user can participate in multiple transactions. The role_type attribute distinguishes between sender and receiver roles. The System_Log table maintains a one-to-many relationship with Transactions, capturing metadata from the original SMS including service_center and readable_date, enabling full traceability of data processing.

Normalization and Data Integrity

The schema follows Third Normal Form to eliminate data redundancy while maintaining query efficiency. Foreign key constraints enforce referential integrity across all relationships.

4. Data Dictionary

User table

```
mysql> desc User;
```

Field	Type	Null	Key	Default	Extra
user_id	int(11)	NO	PRI	NULL	auto_increment
phone_number	varchar(20)	NO	UNI	NULL	
contact_name	varchar(150)	YES		NULL	
user_type	varchar(50)	YES		NULL	
created_at	datetime	YES		CURRENT_TIMESTAMP	

Category Table

```
mysql> desc Category;
```

Field	Type	Null	Key	Default	Extra
category_id	int(11)	NO	PRI	NULL	auto_increment
category_name	varchar(100)	NO	UNI	NULL	
description	varchar(255)	YES		NULL	
created_at	datetime	YES		CURRENT_TIMESTAMP	

Transaction Table

```
mysql> desc Transaction;
```

Field	Type	Null	Key	Default	Extra
transaction_id	int(11)	NO	PRI	NULL	auto_increment
category_id	int(11)	YES	MUL	NULL	
transaction_reference	varchar(100)	NO	UNI	NULL	
amount	decimal(15,2)	NO		NULL	
transaction_type	varchar(50)	YES		NULL	
transaction_status	varchar(50)	YES		Completed	
transaction_date	datetime	NO	MUL	NULL	
raw_sms_body	text	YES		NULL	

Role Table

```
mysql> desc Role;
```

Field	Type	Null	Key	Default	Extra
role_id	int(11)	NO	PRI	NULL	auto_increment
user_id	int(11)	NO	MUL	NULL	
transaction_id	int(11)	NO	MUL	NULL	
role_type	varchar(20)	NO		NULL	
assigned_at	datetime	YES		CURRENT_TIMESTAMP	

System_Log Table

```
mysql> desc System_Log;
```

Field	Type	Null	Key	Default	Extra
log_id	int(11)	NO	PRI	NULL	auto_increment
transaction_id	int(11)	NO	MUL	NULL	
log_status	varchar(50)	YES		NULL	
service_center	varchar(100)	YES		NULL	
readable_date	varchar(100)	YES		NULL	
logged_at	datetime	YES		CURRENT_TIMESTAMP	

5. Sample Queries

- SELECT * FROM User;

```
mysql> SELECT * FROM User;
+-----+-----+-----+-----+-----+
| user_id | phone_number | contact_name | user_type | created_at |
+-----+-----+-----+-----+-----+
| 1 | 250780000001 | Alice Munyaneza | Individual | 2026-01-28 10:20:28 |
| 2 | 250780000002 | Wakuma Kamana | Individual | 2026-01-28 10:20:28 |
| 3 | 250780000003 | Supermarket Ltd | Merchant | 2026-01-28 10:20:28 |
| 4 | 250780000004 | Agent John | Agent | 2026-01-28 10:20:28 |
| 5 | 250780000005 | Charlie Umutoni | Individual | 2026-01-28 10:20:28 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

- SELECT * FROM Category;

```
mysql> SELECT * FROM Category;
+-----+-----+-----+-----+
| category_id | category_name | description | created_at |
+-----+-----+-----+-----+
| 1 | P2P Transfer | Person to person money transfer | 2026-01-28 10:20:28 |
| 2 | Airtime Top-up | Purchase of mobile airtime | 2026-01-28 10:20:28 |
| 3 | Merchant Payment | Payments made to retail stores | 2026-01-28 10:20:28 |
| 4 | Cash Out | Withdrawals at agent locations | 2026-01-28 10:20:28 |
| 5 | Utility Bill | Payments for water, electricity, etc. | 2026-01-28 10:20:28 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

6. Unique rules added to enhance security

Primary Key Constraint

Each table has a unique primary key to ensure every record is uniquely identifiable.

Table name	Primary Key
User	user_id
Transaction	transaction_id
Category	category_id
Role	role_id
System_Log	log_id

Foreign Key Constraint

Foreign keys ensure that relationships between tables remain valid and prevent orphan records.

Table name	Foreign Key	References
Transaction	category_id	Category(category_id)
Role	transaction_id	User(user_id)
Role	transaction_id	Transaction(transaction_id)
System_Log	transaction_id	Transaction(transaction_id)

Timestamp Tracking

All tables include timestamp fields to track when records were created, supporting audit and data tracking.

Table name	Timestamp field
User	created_at
Transaction	transaction_date
Category	created_at
Role	assigned_at
System_Log	logged_at

Screenshots

```
mysql> SELECT tc.CONSTRAINT_NAME, tc.CONSTRAINT_TYPE FROM information_schema.TABLE_CONSTRAINTS tc
-> WHERE tc.TABLE_SCHEMA = 'momo-sms-analytics' AND tc.TABLE_NAME = 'User';
```

```
+-----+-----+
| CONSTRAINT_NAME | CONSTRAINT_TYPE |
+-----+-----+
| PRIMARY         | PRIMARY KEY     |
| phone_number    | UNIQUE          |
+-----+-----+
```

```
mysql> SELECT tc.CONSTRAINT_NAME, tc.CONSTRAINT_TYPE FROM information_schema.TABLE_CONSTRAINTS tc
-> WHERE tc.TABLE_SCHEMA = 'momo-sms-analytics' AND tc.TABLE_NAME = 'Role';
```

```
+-----+-----+
| CONSTRAINT_NAME | CONSTRAINT_TYPE |
+-----+-----+
| PRIMARY         | PRIMARY KEY     |
| fk_role_transaction | FOREIGN KEY     |
| fk_role_user     | FOREIGN KEY     |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT tc.CONSTRAINT_NAME, tc.CONSTRAINT_TYPE FROM information_schema.TABLE_CONSTRAINTS tc
-> WHERE tc.TABLE_SCHEMA = 'momo-sms-analytics' AND tc.TABLE_NAME = 'Transaction';
```

CONSTRAINT_NAME	CONSTRAINT_TYPE
PRIMARY	PRIMARY KEY
transaction_reference	UNIQUE
fk_transaction_category	FOREIGN KEY

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT tc.CONSTRAINT_NAME, tc.CONSTRAINT_TYPE FROM information_schema.TABLE_CONSTRAINTS tc
-> WHERE tc.TABLE_SCHEMA = 'momo-sms-analytics' AND tc.TABLE_NAME = 'Category';
```

CONSTRAINT_NAME	CONSTRAINT_TYPE
PRIMARY	PRIMARY KEY
category_name	UNIQUE

```
2 rows in set (0.01 sec)
```

```
mysql> SELECT tc.CONSTRAINT_NAME, tc.CONSTRAINT_TYPE FROM information_schema.TABLE_CONSTRAINTS tc
-> WHERE tc.TABLE_SCHEMA = 'momo-sms-analytics' AND tc.TABLE_NAME = 'System_Log';
```

CONSTRAINT_NAME	CONSTRAINT_TYPE
PRIMARY	PRIMARY KEY
fk_log_transaction	FOREIGN KEY

```
2 rows in set (0.00 sec)
```